Discussion: claim boundaries and implications

Hypothesis verdicts before interpretation I

We return an explicit verdict on each pre-stated hypothesis (Introduction, H1–H5) before interpreting it, so the contribution is the adjudication itself rather than a single narrative.

- **H1 (formation strategy is the dominant lever) — supported.** The four panels do not separate into a graded four-way ladder. Competence-first selection (**expertise threshold**) is clearly best at 0.037, while representative sortition, single-bloc selection, and random selection cluster around 0.147-0.148 with overlapping intervals. The resolved result is competence-first versus the bottom cluster, not a precise ordering inside that cluster.

Hypothesis verdicts before interpretation II

- **H2 (concentrating correlated error degrades recovery) — unresolved on the baseline grid, then RESOLVED by the composition-coupled confound.** On the baseline generator the representative-vs-single-bloc contrast is design-limited, for a structural reason: that generator never realizes H2's premise, because its judges err independently and ideology shifts only marginal accuracy, so the strategies coincide *by construction* and no regime sweep can fan them out. Once a composition-coupled, marginal-accuracy-preserving error confound supplies the missing channel ([@fig:blocphase]), the contrast resolves cleanly: single-bloc error exceeds representative error in 180/205 matched regimes (paired mean 0.102, 95% CI ± 0.012), the gap widening from 0.000 to 0.112 as coupling rises. The advantage is real but **conditional**: an orthogonal-axis negative control removes it, so we claim "balancing the axis the confound rides on preserves recovery," not a universal sortition law.

Hypothesis verdicts before interpretation III

- **H3 (size is a sampling knob, not guaranteed improvement) — supported only in the bounded sense.** A paired, regime-controlled test resolves a trio-to-six-seat size effect for 3 of the four strategies, and every resolved effect is a tiny increase in error: random selection (+0.015), representative sortition (+0.007), and competence-first selection (+0.004). Single-bloc is within noise. So more experts do not help here, but they also do not create a material penalty; size is essentially neutral at this grid, and the dominant lever is strategy, not size.

Hypothesis verdicts before interpretation IV

- **H4 (error-correlation is measurable, and recovery degrades with it) — diagnostic confirmed; recovery slope resolved by the composition-coupled instrument.** The diagnostic works — the realized correlation tracks the injected coupling (0.019 to 0.123). The *global-injection* tolerance sweep leaves the recovery-vs-correlation slope unresolved (-0.145, 95% CI [-4.335, 0.911] crosses zero), and we now read that as a limitation of that instrument — it couples correlation uniformly to a fixed trio and uses a small grid — rather than as evidence of no effect. The composition-coupled, marginal-accuracy-preserving sweep ([@f ig:blocphase]), with far more observations per point, resolves the recovery half in the affirmative: recovery error rises with within-bloc coupling for the panels that concentrate the confound, and the closed-form Herfindahl account (§Methods) explains the ordering. So the recovery half is a result, not an open question, under a correctly specified confound.

Hypothesis verdicts before interpretation V

- **H5 (the synthetic ranking transfers to a live single-model panel) — rejected.** At the matched three-seat grain the ranking inverts: the rule that is best blind on synthetic data (expertise threshold, 0.037) is the worst under the live `gemma3:4b` panel (0.347). The robust component is that expertise-threshold flip; the live top three are bunched and the evidence is single-model and n-limited, so we report non-transfer of the best-synthetic rule rather than a precise live winner. The suggested caution, scoped to this one companion model: "choose the most expert judges" most cleanly minimized blind-recovery error on judges of *known* accuracy, yet it was the worst rule once "expert" was merely a prompt label the model need not honor — a hypothesis that self-asserted expertise may be an unsafe blind-evaluation selection criterion, worth testing beyond one model rather than an established result.

Hypothesis verdicts before interpretation VI

The remaining subsections elaborate the mechanism behind each verdict.

Practical lesson: selection rule before panel size I

Three usable lessons follow for anyone selecting judges to be evaluated without an answer key. (i) *Which* rule forms the panel matters far more than how many judges it seats: competence-first selection set the lowest blind-recovery error here, while panel size was essentially neutral (H1, H3). (ii) Representative selection protects unsupervised recovery *only* when the lottery balances the very attribute a shared error rides on; balanced on the wrong axis it gives no protection (H2/H4, negative control), and the exposure it controls is a closed-form concentration (Herfindahl) index you can compute on a *proposed* panel before any votes are cast. (iii) The advantage of picking “expert” judges did not survive when judges were a prompted live model rather than parameterized synthetic ones (H5), so a selection rule validated on controllable judges cannot be assumed to carry over to real ones.

Formation strategy is the measured lever I

Studying NTQR *upstream* — at the panel-formation step rather than at the estimator — reframes ground-truth-free evaluation as a selection problem. The strongest finding is that competence-first selection sets a much lower downstream no-answer-key error floor than the other panel-formation rules. The other three strategies cluster tightly enough that their point-estimate order should not be read as a substantive ranking. The scientific claim is therefore about the competence-first-vs-rest separation, not about naming a bottom-cluster winner or loser.

Formation strategy is the measured lever II

That framing matters for application review because peer-review scholarship already treats expert judgment as socially situated and panel-dependent rather than mechanically objective Lamont (2009); Lee et al. (2013), and because empirical studies find substantial reviewer disagreement on the same submitted work Cole et al. (1981); Pier et al. (2018), score-model uncertainty large enough to alter the implied funded set Johnson (2008), and limited grant-productivity predictiveness of NIH percentiles Fang et al. (2016). The manuscript's increment is narrower: it instruments one selection mechanism and asks whether different panel draws change an unlabeled evaluator's oracle-referenced error. The claim is about generated artifacts and one local Gemma stress test, not about the global reliability of academic review.

Formation strategy is the measured lever III

This is, in part, a result *against* the intuitive case for sortition. A representative lottery is the fair, auditable way to form a panel, but on this instrument it does not minimize oracle-referenced EIE error — competence-first does. We report that plainly rather than engineering a narrative in which sortition wins. That is not a general refutation of sortition, deliberative participation, or the "diversity can beat ability" result. Those arguments rely on different objectives and premises: democratic legitimacy and public consultation Fishkin (2009), search diversity Hong and Page (2004), and jury-theorem aggregation under competence and conditional-independence assumptions Grofman et al. (1983). The present result is narrower: in this binary noisy-judge instrument, competence-first sampling gives the lowest oracle-referenced EIE error. Relative to the single-bloc comparator the representative draw's point estimate is now reported over the full active regime grid rather than collapsed to two panel-size means. Some cells resolve in the predicted direction, others remain descriptive or design-limited, so we claim artifact-bounded regime structure, not a general sortition advantage over single-bloc selection

Design-limited nulls remain results I

Two results are bounded rather than universal, and we do not dress them up.

1. **On the baseline grid, representative vs ideological is design-limited.** Varying expert stringency, bias spread, and panel size jointly, the heatmap reports which regenerated synthetic cells align with the directional prediction, which resolve by descriptive intervals, and which remain uncertain — but the pooled contrast is not resolved there, *by construction*, because the baseline judges err independently. It resolves only once the composition-coupled confound supplies the correlation channel (see the H2 verdict and [@fig:blocphase]); the null is a property of the baseline design, not of sortition.

Design-limited nulls remain results II

2. **Size is not a uniform power knob.** A paired, regime-controlled contrast resolves a trio-to-six-seat size change for 3 of the four strategies, and each resolved change is a small increase in error. The largest delta is $+0.015$, so more experts do not help and at most very slightly hurt. The clean "more experts always helps" story is rejected, but the result is essentially neutral rather than a material size penalty. That is consistent with peer-review jury-theorem work: adding reviewers helps only under assumptions about competence, dependence, and aggregation that must be checked rather than presumed Arvan et al. (2025).

Reporting these nulls is the point of building a measurement instrument rather than a demonstration.

Independence explains why strategy ordering changes I

NTQR's EIE solver rests on the judges' errors being approximately independent. On the baseline generator, single-bloc selection is indistinguishable from representative and random selection **by construction**: ideology there shifts only each judge's marginal accuracy, every judge errs from an independent stream, and an agreement-only estimator cannot be moved by composition. Single-bloc becomes the *separable* adversarial comparator only once the composition-coupled confound supplies a genuine cross-judge error-correlation channel ([@fig:blocphase]). Competence-first panels pair high accuracy with whatever independence the population affords, giving the solver the easiest system to invert. The fair-lottery argument should therefore be made from auditability, representation, and bounded empirical performance rather than from an unqualified error-minimization win.

Error independence must be measured before interpretation I

Error independence must be measured before interpretation II

The assumption the whole ordering hangs on — that judges' errors are approximately independent — is, in this instrument, no longer an assumption but a measured quantity. The controlled-correlation sweep confirms the *knob works*: the realized pairwise correlation NTQR reports rises with the injected coupling (0.019 to 0.123). What it does **not** yet resolve, at this grid, is whether that correlation degrades recovery — the fitted slope is **statistically indistinguishable from zero** (-0.145, 95% CI [-4.335, 0.911] spans zero), unresolved rather than absent — and the power layer explains why that is unsurprising rather than disappointing: 12 of 28 contrasts are well-powered at the current seed count (MDE 0.101), 13 reach nominal significance, and 12 survive Holm correction. The honest reading is that the current design resolves the largest separations but leaves smaller neighboring contrasts design-limited. The strategy *ranking* remains an ordering of point estimates; the analysis says exactly how many seeds would let the unresolved contrasts resolve. That global-injection slope is, in any

Scholarship frames the stress test, not the evidence level I

The postdoctoral-review setting is intentionally close to a literature where selection, status, and bias are known concerns. Cumulative-advantage accounts of scientific recognition Merton (1968), resubmission experiments showing fragility in journal review Peters and Ceci (1982), blind-review experiments and observational bias studies Tomkins et al. (2017); Helmer et al. (2017), and empirical studies of fellowship or grant outcomes Wennerås and Wold (1997); Ginther et al. (2011) make it reasonable to study reviewer sampling, not only evaluator algebra. The age axis has the same status: ageism and age-discrimination findings motivate it as a protected-attribute stress test North and Fiske (2013); Neumark et al. (2019), but the manuscript does not infer anything about real postdoctoral age discrimination from prompted Gemma votes.

Scholarship frames the stress test, not the evidence level II

The synthetic and Gemma tracks therefore answer different questions. The synthetic track can make controlled claims because it owns the oracle label and the expert parameters. The live Gemma track can only show whether the same sampling vocabulary produces measurable traces under one local model with serialized provenance. Prompted LLM evaluation is itself an active measurement problem, not a neutral readout Zheng et al. (2023), and language-model risk scholarship cautions against treating model text as a transparent substitute for human judgment Bender et al. (2021), so the correct inference level is empirical feasibility plus directional stress testing. Lottery and collective-allocation proposals in science funding Bollen et al. (2014); Fang and Casadevall (2016), and maverick-science arguments for lotteries Avin (2019) make randomized institutional design a legitimate comparator, but they do not license a claim that lottery-formed reviewer panels optimize NTQR recovery. Scholarship supplies the problem context and the variables worth stress-testing; regenerated artifacts supply the evidence.

Limitations: synthetic scope, single-model live evidence, historical analogy I

Limitations: synthetic scope, single-model live evidence, historical analogy II

The reported `manuscript_contrast` grid fixes prevalence and corpus size while varying mean expertise, bias, panel size, and strategy; it uses 96 experts, 300 as the modal item count in the rendered tokens, and up to 8 trios per panel over 96 seeds. The repository now defines broader sensitivity and finer panel-ladder profiles, but those profiles are configuration surfaces until regenerated and audited as manuscript evidence; the reported results remain bounded to the active profile. The oracle-closest tie-break is deliberately charitable to the unsupervised estimate, so reported errors are a lower bound on what a blind tie-break would incur. The alarm's $O(Q^3)$ cost confines the consistency-alarm track to small corpora ($Q \leq 30$), so the alarm cannot yet serve as a sweep-scale signal. The statistical-power analysis shows most strategy contrasts are underpowered at the current seed count, so several headline comparisons are design-limited rather than settled. The Gemma postdoctoral panel is also bounded: it uses fictitious applications, synthetic age metadata, prompted reviewer personas,

Synthetic and live tracks operate at different inference levels I

This manuscript follows a standard division between controlled experiment and empirical companion evidence. The deterministic synthetic track is the controlled Results spine: it generates the strategy-ranking, panel-size, controlled-correlation, power-budget, alarm-cost, and analytical-alignment numbers from regenerated local artifacts. Those claims are validated against known oracle labels because the generator owns the truth labels.

The real-Ollama track is reported as separate live artifacts and empirical companion evidence, not a pooled extension of the synthetic sweep. It was performed locally with required-live `gemma3:4b` (full provenance in Methods), showing the same sampling vocabulary run on a real local model prompted as different reviewers. It does not validate the full synthetic regime grid, establish a population effect size, or prove that Gemma substitutes for human reviewers.

Synthetic and live tracks operate at different inference levels II

The combined interpretation is therefore deliberately tiered: synthetic experiments support the controlled mechanism and regime maps; analytical checks test directional expectations against those regenerated artifacts; and live Ollama runs demonstrate empirical feasibility plus n-limited directional support. Wider parameter sweeps and larger empirical panels are the next steps before any stronger general claim.

Data, code, and generated-artifact availability I

Data, code, and generated-artifact availability II

All source code, methods, and documentation are openly available at the public repository `docxology/ntqr_allotment`: the deterministic synthetic instrument, the bloc-confound and Herfindahl modules, the live Gemma vote cache with serialized model provenance, every figure, and the manuscript regeneration pipeline. Every reported number is token-injected from `output/data/` by `src/ntqr_allotment/manuscript_variables.py`, so no result is hand-transcribed and the manuscript regenerates from source under a zero-orphan token contract. The synthetic track is fully deterministic under fixed seeds (profile config hash `fda4da941cf0`); the live track reproduces against a local Ollama `gemma3:4b` instance (digest `a2af6cc3eb7f`, config hash `5161ffe474b3`) using the serialized, resumable per-vote cache keyed on the config hash, seed, reviewer, application, model digest, and decode parameters. A steganographic provenance variant of the PDF additionally carries an extractable hash of the current source PDF, verified by `scripts/verify_stego.py`. The archival DOI is 10.5281/zenodo.21083779.

Ethics, protected attributes, and competing interests I

The postdoctoral-review setting is entirely synthetic: the applications, latent quality labels, reviewer personas, and age metadata are all generated, and no human subjects, real applicants, or real review records are involved. True latent quality is generated independently of age; age enters only as a protected-attribute stress test for bias expression under sampling, and its use here is diagnostic, not an endorsement of using age in real admissions, hiring, or fellowship review. The live language-model outputs are treated as an instrumented measurement of a prompted system, not as a substitute for human judgment. The authors declare no competing interests.